

Project: FoodBridge

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Selected SDG and Reason for Selection

Selected SDG: Goal 2: Zero Hunger

What the SDG focuses on: Goal 2 aims to end hunger, achieve food security, improve nutrition, and promote sustainable agriculture. It focuses on ensuring that all people, especially the poor and vulnerable, have access to safe, nutritious, and sufficient food all year round.

Why this issue is important: Globally, about one-third of all food produced for human consumption is lost or wasted, while millions of people go hungry every day. Addressing this imbalance is critical not only for human survival and dignity but also for environmental sustainability, as food waste is a major contributor to greenhouse gas emissions.

Problem Statement

What is the problem? There is a massive disconnect between surplus food generated by local businesses (restaurants, supermarkets, bakeries) and the local shelters and NGOs struggling to feed vulnerable populations.

Where does this problem exist? This issue exists primarily in urban and suburban areas where commercial food operations produce daily surplus, yet local shelters face food shortages due to lack of real-time communication and logistics.

Who is affected? It affects low-income families, homeless individuals, and anyone relying on food banks or shelters.

Why is it serious? It is serious because throwing away edible food while people starve nearby is a logistical and moral failure. It wastes resources (water, labor, land) used to produce the food and deprives people of basic sustenance.

What could happen if it is not solved? If left unsolved, food insecurity will continue to rise, leading to malnutrition, poor health outcomes, and increased environmental degradation from rotting food in landfills.

Proposed Solution

What the project is: FoodBridge is a full-stack web application designed to connect food donors directly with local NGOs and shelters.

How the system works: Donors (restaurants or grocery stores) create an account and can instantly post details about surplus food (type, quantity, expiry time, pickup location). NGOs can log in to their dashboard, see a real-time feed of available food in their area, and click a button to "Claim" the food, which updates the status and notifies the donor.

How it helps solve the problem: It eliminates the communication gap by providing a centralized, real-time platform where supply (surplus food) can immediately meet demand (shelters needing food).

Who will use the application:

1. **Donors:** Restaurant managers, grocery store owners, event organizers.
2. **Receivers:** NGO volunteers, shelter coordinators, food bank managers.

Project Features

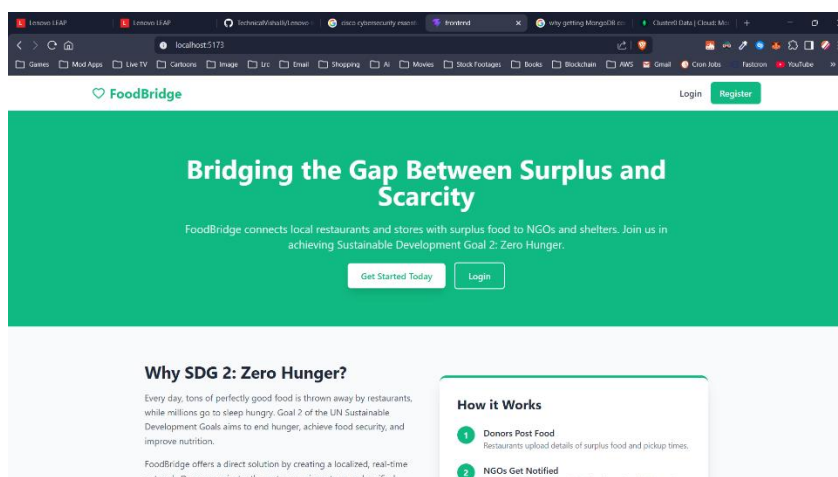
1. **Role-Based Authentication:** Secure login and registration tailored for either Donors or NGOs.
2. **Donor Dashboard:** An interface for donors to easily post new food listings and track their donation history.
3. **Real-time Claiming System:** A live feed where NGOs can view available donations and claim them before they expire.
4. **Interactive UI:** A responsive, modern design ensuring usability across desktop and mobile devices.

Technology Stack

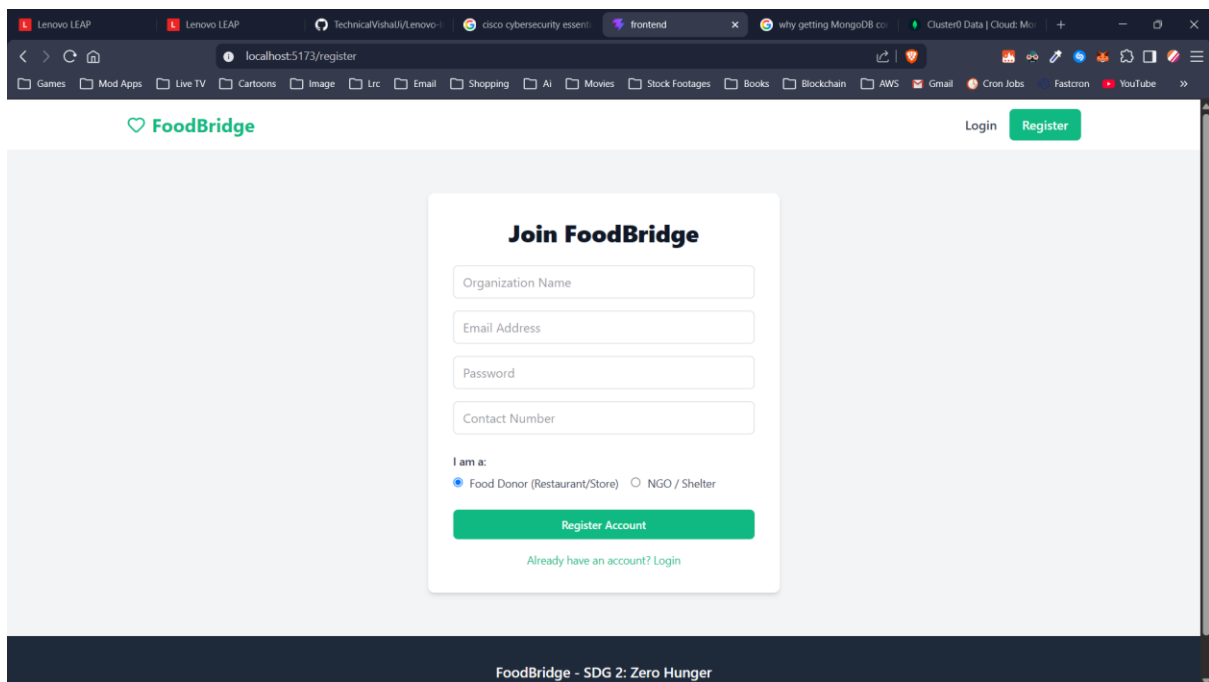
- **Frontend:** ReactJS, TailwindCSS, React Router DOM, Vite
- **Backend:** Node.js, Express.js
- **Database:** MongoDB (with Mongoose ORM)
- **Authentication:** JSON Web Tokens (JWT), bcryptjs

Project Screenshots (Proof of Work)

Homepage:



Registration:



The screenshot shows a web browser window with the URL `localhost:5173/register`. The page features the FoodBridge logo (a green heart icon followed by the text "FoodBridge") in the top left and "Login" and "Register" buttons in the top right. The main content area is a light gray rectangle containing a white registration form titled "Join FoodBridge". The form includes four input fields: "Organization Name", "Email Address", "Password", and "Contact Number". Below these fields is a section labeled "I am a:" with two radio button options: "Food Donor (Restaurant/Store)" (which is selected) and "NGO / Shelter". At the bottom of the form is a green "Register Account" button and a link that says "Already have an account? Login". The footer of the page is dark gray with the text "FoodBridge - SDG 2: Zero Hunger".

Join FoodBridge

Organization Name

Email Address

Password

Contact Number

I am a:

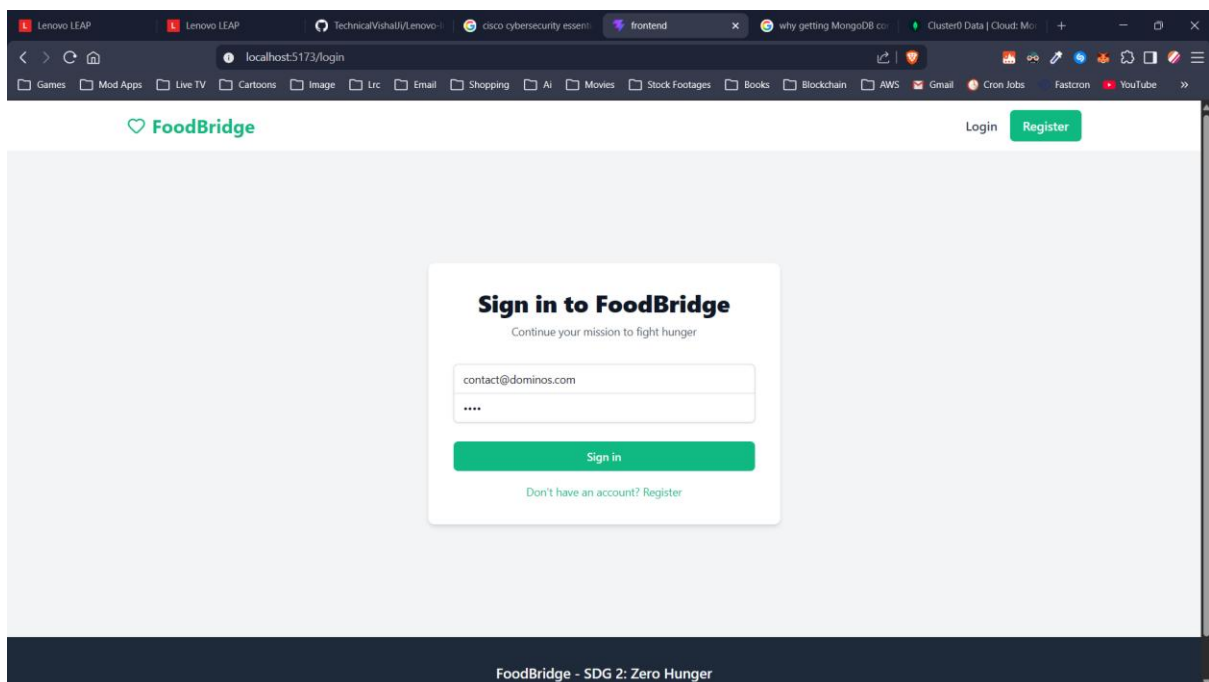
☒ Food Donor (Restaurant/Store) ☐ NGO / Shelter

Register Account

Already have an account? Login

FoodBridge - SDG 2: Zero Hunger

Login:



The screenshot shows a web browser window with the URL `localhost:5173/login`. The page features the FoodBridge logo (a green heart icon followed by the text "FoodBridge") in the top left and "Login" and "Register" buttons in the top right. The main content area is a light gray rectangle containing a white login form titled "Sign in to FoodBridge". Below the title is a subtitle "Continue your mission to fight hunger". The form includes two input fields: "contact@dominos.com" and "****". At the bottom of the form is a green "Sign in" button and a link that says "Don't have an account? Register". The footer of the page is dark gray with the text "FoodBridge - SDG 2: Zero Hunger".

Sign in to FoodBridge

Continue your mission to fight hunger

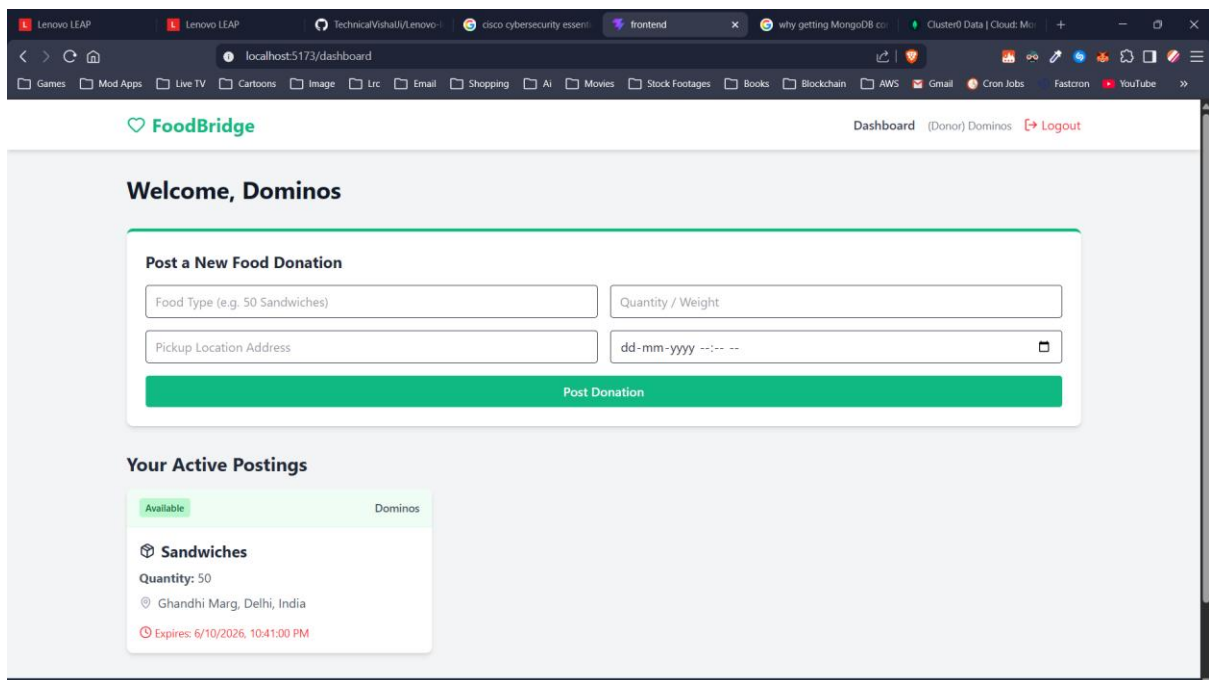
contact@dominos.com

Sign in

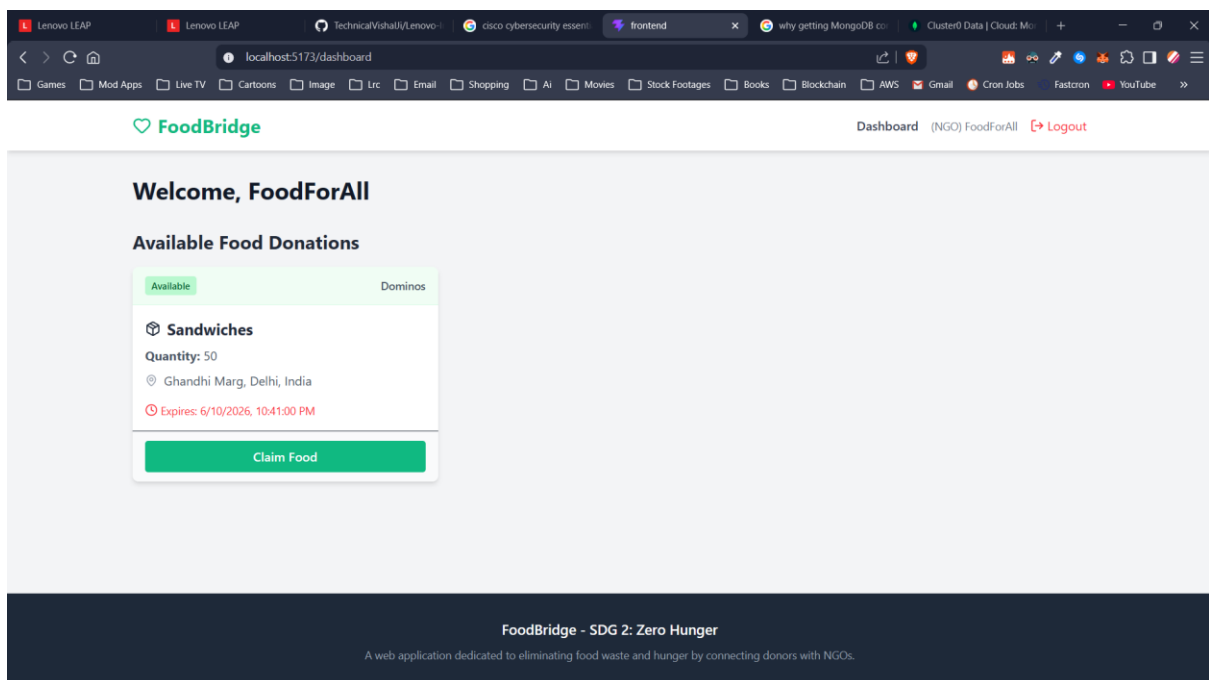
Don't have an account? Register

FoodBridge - SDG 2: Zero Hunger

Donor Dashboard:



NGO Dashboard:



Project Links

GitHub Repository: [Link](#)

Future Scope

- **Mobile Application:** Developing a React Native app for push notifications so NGOs can be instantly alerted when food is posted.
- **Delivery Integration:** Partnering with volunteer driver networks (like DoorDash for Good) to handle the transportation of food automatically.
- **Data Analytics:** Providing donors with tax-deductible receipt generation and impact reports showing exactly how many meals they have provided.